

Anesthesia for Immune Mediated and Infectious Diseases MCQ – Chapter 25

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 25)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the underlying pathology of rheumatoid arthritis?

- A. Immune-mediated vasculitis
- B. Immune-mediated synovitis
- C. Degenerative joint disease
- D. Autoimmune myositis

Ans: B

Q2. Which associated comorbidities should be ruled out in rheumatoid arthritis patients preoperatively?

- A. Renal failure, hepatitis, diabetes
- B. Cardiac valvular lesions, pericarditis, and pulmonary fibrosis
- C. Thyroid dysfunction and adrenal insufficiency
- D. Peptic ulcer and esophageal varices

Ans: B

Q3. What is the most important concern in anesthetic management of rheumatoid arthritis?

- A. Drug interactions
- B. Difficult airway
- C. Renal dysfunction
- D. Coagulopathy

Ans: B

Q4. Which side effects should be considered preoperatively in rheumatoid arthritis patients?

- A. Side effects of antihypertensives
- B. Side effects of DMARDs, NSAIDs, and corticosteroids
- C. Side effects of antibiotics
- D. Side effects of anticoagulants only

Ans: B

Q5. What is the preferred choice of anesthesia in rheumatoid arthritis patients?

- A. General anesthesia
- B. Regional anesthesia
- C. Local anesthesia only
- D. Monitored anesthesia care

Ans: B

Q6. Why is the lumbar spine usually suitable for spinal/epidural in rheumatoid arthritis?

- A. It has better blood supply
- B. Lumbar spine is usually not involved in rheumatoid arthritis
- C. It is easier to access
- D. It provides better block

Ans: B

Q7. Why is general anesthesia difficult and risky in rheumatoid arthritis patients?

- A. Due to drug resistance
- B. Due to difficult intubation
- C. Due to cardiac instability
- D. Due to renal dysfunction

Ans: B

Q8. What can happen if the neck is flexed in patients with atlantoaxial subluxation? A. Cervical disc herniation B. Odontoid process can compress spinal cord leading to quadriplegia C. Vertebral artery dissection D. Esophageal injury	Ans: B
Q9. How should intubation be performed in rheumatoid arthritis patients with atlantoaxial subluxation? A. Rapid sequence intubation with neck flexion B. Awake intubation under topical anesthesia, or under GA with neck in extension C. Blind nasal intubation D. Tracheostomy	Ans: B
Q10. What are the three causes of difficult intubation in rheumatoid arthritis? A. Macroglossia, prognathism, epiglottic overgrowth B. Atlantoaxial instability, stiffness of cervical spine, temporomandibular and cricoarytenoid arthritis C. Tracheal stenosis, vocal cord paralysis, subglottic edema D. Obesity, short neck, large tongue	Ans: B
Q11. Why may arterial cannulation be difficult in rheumatoid arthritis patients? A. Due to small vessel caliber B. Due to calcified radial arteries C. Due to vasospasm D. Due to thrombosed vessels	Ans: B
Q12. Why may central venous cannulation be difficult in rheumatoid arthritis? A. Due to thrombosis B. Due to cervical spine fusion C. Due to superior vena cava obstruction D. Due to coagulopathy	Ans: B
Q13. Why can ankylosing spondylitis be the most difficult case from anesthesia point of view? A. Due to cardiac involvement B. Fusion of lumbar spine makes CNB impossible and fusion of cervical spine makes intubation impossible C. Due to renal involvement D. Due to coagulopathy	Ans: B
Q14. How should intubation be performed in ankylosing spondylitis patients with cervical spine fusion? A. Direct laryngoscopy under GA B. Awake state under fiberoptic bronchoscopy C. Blind nasal intubation D. Rapid sequence intubation	Ans: B

Q15. Which cardiac conditions are common in ankylosing spondylitis? A. Mitral stenosis and atrial fibrillation B. Aortic regurgitation and bundle branch block C. Ventricular septal defect D. Pericardial effusion	Ans: B
Q16. What determines the perioperative morbidity in systemic lupus erythematosus (SLE)? A. Duration of disease B. Involvement of major organ functions C. Age of the patient D. Type of surgery	Ans: B
Q17. What laryngeal involvement in SLE can make intubation difficult? A. Laryngeal edema only B. Cricoarytenoid arthritis and laryngeal nerve palsy C. Vocal cord polyps D. Subglottic stenosis	Ans: B
Q18. Which organ involvement in scleroderma significantly affects perioperative morbidity? A. Renal failure and hepatic dysfunction B. Pulmonary fibrosis and pulmonary hypertension C. Cardiac tamponade D. Cerebrovascular disease	Ans: B
Q19. Why can intubation be difficult in scleroderma? A. Due to cervical spine fusion B. Due to small oral aperture C. Due to macroglossia D. Due to tracheal compression	Ans: B
Q20. Why can IV cannulation be difficult in scleroderma? A. Due to vasospasm B. Due to dermal thickening C. Due to edema D. Due to calcification of vessels	Ans: B
Q21. How are HIV positive patients classified according to ASA grading? A. ASA I B. ASA II C. ASA III D. ASA IV	Ans: B

Q22. How are patients with AIDS classified according to ASA grading? A. ASA I/II B. ASA II/III C. ASA III/IV D. ASA IV/V	Ans: C
Q23. Which side effects of HAART should be ruled out preoperatively? A. Hypothyroidism and osteoporosis B. Diabetes, coronary and cerebrovascular disease, deranged liver function C. Renal failure only D. Anemia only	Ans: B
Q24. Patients who started HAART within how many months of surgery suffer more from perioperative complications? A. 3 months B. 6 months C. 9 months D. 12 months	Ans: B
Q25. Do CD4 count and viral load increase perioperative anesthetic complications? A. Yes, significantly B. Yes, but only in emergency surgery C. No, they affect overall prognosis but not perioperative anesthetic complications D. Only viral load increases complications	Ans: C
Q26. Can regional anesthesia be given to HIV/AIDS patients? A. Never, it is contraindicated B. Yes, safely if there are no associated neuropathies C. Only epidural is safe D. Only spinal is safe	Ans: B
Q27. What can make intubation difficult in HIV/AIDS patients? A. Cervical spine fusion B. Fat redistribution around the neck C. Tracheal compression D. Vocal cord paralysis	Ans: B
Q28. What is vital during any procedure in HIV/AIDS patients? A. Use of specialized equipment B. Maintaining universal precautions and high degree of asepsis C. Avoiding all regional blocks D. Using only disposable equipment	Ans: B

Q29. In rheumatoid arthritis, intubation should be performed with the neck in which position? A. Neutral position B. Flexion C. Extension D. Lateral rotation	Ans: C
Q30. Which joint involvement in rheumatoid arthritis contributes to difficult intubation besides atlantoaxial and cervical spine? A. Shoulder and elbow joints B. Temporomandibular and cricoarytenoid joints C. Hip and knee joints D. Sacroiliac joints	Ans: B
Q31. Which of the following is TRUE about ankylosing spondylitis and anesthesia? A. Spinal anesthesia is always easy B. General anesthesia with direct laryngoscopy is preferred C. It may turn out to be the most difficult case from anesthesia point of view D. Epidural is the preferred technique	Ans: C
Q32. In ankylosing spondylitis, fusion of which spine makes central neuraxial blocks impossible? A. Cervical spine B. Thoracic spine C. Lumbar spine D. Sacral spine	Ans: C
Q33. In ankylosing spondylitis, fusion of which spine makes intubation impossible? A. Lumbar spine B. Thoracic spine C. Cervical spine D. Sacral spine	Ans: C
Q34. What type of valvular disease is commonly associated with ankylosing spondylitis? A. Mitral stenosis B. Mitral regurgitation C. Aortic regurgitation D. Aortic stenosis	Ans: C
Q35. What conduction abnormality is common in ankylosing spondylitis? A. Atrial fibrillation B. Bundle branch block C. Complete heart block D. Ventricular tachycardia	Ans: B

Q36. Which condition is characterized by immune-mediated synovitis? A. Ankylosing spondylitis B. Systemic lupus erythematosus C. Rheumatoid arthritis D. Scleroderma	Ans: C
Q37. In which condition can small oral aperture cause difficult intubation? A. Rheumatoid arthritis B. Ankylosing spondylitis C. Scleroderma D. SLE	Ans: C
Q38. Which condition can cause cricoarytenoid arthritis leading to difficult intubation — BOTH rheumatoid arthritis AND SLE? A. True – both RA and SLE can cause cricoarytenoid arthritis B. False – only RA causes it C. False – only SLE causes it D. False – neither causes it	Ans: A
Q39. Which is the best method of intubation in a patient with atlantoaxial subluxation under general anesthesia? A. Direct laryngoscopy with neck flexion B. Intubation with neck in extension C. Blind nasal intubation D. Laryngeal mask airway	Ans: B
Q40. Patients with atlantoaxial subluxation cannot tolerate which neck movement? A. Extension B. Flexion C. Lateral rotation D. All movements	Ans: B
Q41. What is HAART in the context of HIV management? A. Highly Active Antimicrobial Therapy B. Highly Active Antiretroviral Therapy C. High Altitude Antiviral Regimen Therapy D. Hepatic Active Antiretroviral Treatment	Ans: B
Q42. Which pulmonary conditions in scleroderma affect perioperative outcome? A. Bronchiectasis and emphysema B. Pulmonary fibrosis and pulmonary hypertension C. Pleural effusion and pneumothorax D. Asthma and COPD	Ans: B

Q43. What is the awake intubation technique of choice for ankylosing spondylitis with cervical spine fusion? A. Direct laryngoscopy B. Video laryngoscopy C. Fiberoptic bronchoscopy D. Blind nasal intubation	Ans: C
Q44. Why are HIV positive patients classified as at least ASA II? A. Due to high risk of bleeding B. Due to involvement of multiple organs C. Due to immunosuppression only D. Due to drug interactions	Ans: B
Q45. Which condition among the immune mediated diseases can cause laryngeal nerve palsy? A. Rheumatoid arthritis B. Ankylosing spondylitis C. Systemic lupus erythematosus D. Scleroderma	Ans: C
Q46. What is the cause of fat redistribution around the neck in HIV/AIDS patients? A. The disease itself B. Side effect of HAART (lipodystrophy) C. Malnutrition D. Opportunistic infections	Ans: B
Q47. Which immune mediated disease most commonly involves the cervical spine causing intubation difficulty? A. SLE B. Scleroderma C. Rheumatoid arthritis and Ankylosing spondylitis D. HIV/AIDS	Ans: C
Q48. In which condition should neuropathies be ruled out before giving regional anesthesia? A. Scleroderma B. HIV/AIDS C. Ankylosing spondylitis D. Rheumatoid arthritis	Ans: B
Q49. Pericarditis is an associated comorbidity of which immune mediated disease? A. Ankylosing spondylitis B. Scleroderma C. Rheumatoid arthritis D. HIV/AIDS	Ans: C

Q50. Pulmonary fibrosis is a comorbidity associated with which TWO immune mediated diseases from this chapter?

- A. RA and SLE
- B. Rheumatoid arthritis and Scleroderma
- C. SLE and Scleroderma
- D. Ankylosing spondylitis and Scleroderma

Ans: B

Answer Key & Explanations

Q1. Answer: B — Immune-mediated synovitis

Rheumatoid arthritis is a disease characterized by immune-mediated synovitis.

Topic: Rheumatoid Arthritis | Ref: Ch 25, Page 214

Q2. Answer: B — Cardiac valvular lesions, pericarditis, and pulmonary fibrosis

Associated comorbidities like cardiac valvular lesions, pericarditis, and pulmonary fibrosis should be ruled out.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

Q3. Answer: B — Difficult airway

As difficult airway is the most important concern, airway evaluation should be done very meticulously.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

Q4. Answer: B — Side effects of DMARDs, NSAIDs, and corticosteroids

Side effects of disease modifying antirheumatic drugs (DMARDs), NSAIDs and corticosteroids should be taken into consideration.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

Q5. Answer: B — Regional anesthesia

Regional anesthesia is the choice. Lumbar spine is usually not involved in rheumatoid arthritis therefore spinal/epidural can be given easily.

Topic: Rheumatoid Arthritis – Choice of Anesthesia | Ref: Ch 25, Page 214

Q6. Answer: B — Lumbar spine is usually not involved in rheumatoid arthritis

Lumbar spine is usually not involved in rheumatoid arthritis therefore spinal/epidural can be given easily.

Topic: Rheumatoid Arthritis – Choice of Anesthesia | Ref: Ch 25, Page 214

Q7. Answer: B — Due to difficult intubation

General anesthesia is difficult and risky due to difficult intubation caused by atlantoaxial instability, stiffness of cervical spine, and temporomandibular and cricoarytenoid arthritis.

Topic: Rheumatoid Arthritis – General Anesthesia | Ref: Ch 25, Page 214

Q8. Answer: B — Odontoid process can compress spinal cord leading to quadriplegia

Flexion of neck can cause odontoid process to compress spinal cord leading to quadriplegia.

Topic: Rheumatoid Arthritis – Atlantoaxial Instability | Ref: Ch 25, Page 214

Q9. Answer: B — Awake intubation under topical anesthesia, or under GA with neck in extension

Intubation is best performed under topical anesthesia with awake patient. If under GA, intubation should be done with neck in extension.

Topic: Rheumatoid Arthritis – Intubation | Ref: Ch 25, Page 214

Q10. Answer: B — Atlantoaxial instability, stiffness of cervical spine, temporomandibular and cricoarytenoid arthritis

Difficult intubation is due to atlantoaxial instability, stiffness of cervical spine, and temporomandibular and cricoarytenoid arthritis.

Topic: Rheumatoid Arthritis – General Anesthesia | Ref: Ch 25, Page 214

Q11. Answer: B — Due to calcified radial arteries

Arterial cannulation may be difficult due to calcified radial arteries.

Topic: Rheumatoid Arthritis – Vascular Access | Ref: Ch 25, Page 214

Q12. Answer: B — Due to cervical spine fusion

Central venous cannulation may be difficult due to cervical spine fusion.

Topic: Rheumatoid Arthritis – Vascular Access | Ref: Ch 25, Page 214

Q13. Answer: B — Fusion of lumbar spine makes CNB impossible and fusion of cervical spine makes intubation impossible

Fusion of lumbar spine may make central neuraxial blocks impossible and fusion of cervical spine may make intubation impossible.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q14. Answer: B — Awake state under fiberoptic bronchoscopy

Intubation in such patients should be performed in awake state under fiberoptic bronchoscopy.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q15. Answer: B — Aortic regurgitation and bundle branch block

Aortic regurgitation and bundle branch block are common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q16. Answer: B — Involvement of major organ functions

The involvement of major organ functions determines the perioperative morbidity in SLE.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

Q17. Answer: B — Cricoarytenoid arthritis and laryngeal nerve palsy

Laryngeal involvement including cricoarytenoid arthritis and laryngeal nerve palsy can make intubation difficult in SLE.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

Q18. Answer: B — Pulmonary fibrosis and pulmonary hypertension

Multiorgan involvement, particularly pulmonary fibrosis and pulmonary hypertension can significantly affect perioperative morbidity.

Topic: Scleroderma | Ref: Ch 25, Page 214

Q19. Answer: B — Due to small oral aperture

Small oral aperture can make intubation difficult in scleroderma.

Topic: Scleroderma | Ref: Ch 25, Page 214

Q20. Answer: B — Due to dermal thickening

Dermal thickening can make intravenous cannulation difficult in scleroderma.

Topic: Scleroderma | Ref: Ch 25, Page 214

Q21. Answer: B — ASA II

Due to involvement of multiple organs all HIV positive patients are categorized as ASA grade II.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q22. Answer: C — ASA III/IV

Patients with AIDS are categorized as ASA III/IV.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q23. Answer: B — Diabetes, coronary and cerebrovascular disease, deranged liver function

Side effects of HAART including diabetes, coronary and cerebrovascular disease, deranged liver function must be ruled out preoperatively.

Topic: HIV and AIDS – Preoperative Evaluation | Ref: Ch 25, Page 215

Q24. Answer: B — 6 months

Patients who have started HAART within 6 months of surgery suffer more from perioperative complications.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q25. Answer: C — No, they affect overall prognosis but not perioperative anesthetic complications

CD4 count and viral load do affect the overall prognosis after surgery but have not been found to increase perioperative anesthetic complications.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q26. Answer: B — Yes, safely if there are no associated neuropathies

Regional anesthesia can be given safely if there are no associated neuropathies.

Topic: HIV and AIDS – Choice of Anesthesia | Ref: Ch 25, Page 215

Q27. Answer: B — Fat redistribution around the neck

Fat redistribution around the neck can make intubation difficult in HIV/AIDS patients.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q28. Answer: B — Maintaining universal precautions and high degree of asepsis

Maintaining universal precautions and high degree of asepsis during any procedure is vital.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q29. Answer: C — Extension

Due to atlantoaxial instability, intubation in RA patients should be performed with neck in extension.

Topic: Rheumatoid Arthritis – Key Points | Ref: Ch 25, Page 214

Q30. Answer: B — Temporomandibular and cricoarytenoid joints

Temporomandibular and cricoarytenoid arthritis contribute to difficult intubation in rheumatoid arthritis.

Topic: Rheumatoid Arthritis – Difficult Intubation | Ref: Ch 25, Page 214

Q31. Answer: C — It may turn out to be the most difficult case from anesthesia point of view

From anesthesia point of view ankylosing spondylitis may turn out to be the most difficult case at times.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q32. Answer: C — Lumbar spine

Fusion of lumbar spine may make central neuraxial blocks impossible.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q33. Answer: C — Cervical spine

Fusion of cervical spine may make intubation impossible in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q34. Answer: C — Aortic regurgitation

Aortic regurgitation is common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q35. Answer: B — Bundle branch block

Bundle branch block is common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q36. Answer: C — Rheumatoid arthritis

Rheumatoid arthritis is a disease characterized by immune-mediated synovitis.

Topic: Rheumatoid Arthritis | Ref: Ch 25, Page 214

Q37. Answer: C — Scleroderma

In scleroderma, small oral aperture can make intubation difficult.

Topic: Scleroderma | Ref: Ch 25, Page 214

Q38. Answer: A — True – both RA and SLE can cause cricoarytenoid arthritis

Both rheumatoid arthritis (temporomandibular and cricoarytenoid arthritis) and SLE (laryngeal involvement including cricoarytenoid arthritis) can cause difficult intubation.

Topic: RA and SLE – Airway Management | Ref: Ch 25, Page 214

Q39. Answer: B — Intubation with neck in extension

If intubation has to be done under general anesthesia, it should be done with neck in extension to avoid odontoid compression of spinal cord.

Topic: Rheumatoid Arthritis – Intubation | Ref: Ch 25, Page 214

Q40. Answer: B — Flexion

Patients of atlantoaxial subluxation cannot tolerate flexion as it can cause odontoid process to compress spinal cord.

Topic: Rheumatoid Arthritis – Atlantoaxial Instability | Ref: Ch 25, Page 214

Q41. Answer: B — Highly Active Antiretroviral Therapy

HAART stands for Highly Active Antiretroviral Therapy used in HIV management.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q42. Answer: B — Pulmonary fibrosis and pulmonary hypertension

Pulmonary fibrosis and pulmonary hypertension in scleroderma significantly affect perioperative morbidity.

Topic: Scleroderma | Ref: Ch 25, Page 214

Q43. Answer: C — Fiberoptic bronchoscopy

Intubation should be performed in awake state under fiberoptic bronchoscopy in ankylosing spondylitis patients.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q44. Answer: B — Due to involvement of multiple organs

Due to involvement of multiple organs all HIV positive patients are categorized as ASA grade II.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q45. Answer: C — Systemic lupus erythematosus

In SLE, laryngeal involvement including laryngeal nerve palsy can make intubation difficult.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

Q46. Answer: B — Side effect of HAART (lipodystrophy)

Fat redistribution (lipodystrophy) is a known side effect of HAART which can cause fat accumulation around the neck making intubation difficult.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

Q47. Answer: C — Rheumatoid arthritis and Ankylosing spondylitis

Both RA (atlantoaxial instability, cervical spine stiffness) and ankylosing spondylitis (cervical spine fusion) commonly involve the cervical spine causing difficult intubation.

Topic: RA and Ankylosing Spondylitis | Ref: Ch 25, Page 214

Q48. Answer: B — HIV/AIDS

In HIV/AIDS patients, regional anesthesia can be given safely if there are no associated neuropathies.

Topic: HIV and AIDS – Choice of Anesthesia | Ref: Ch 25, Page 215

Q49. Answer: C — Rheumatoid arthritis

Associated comorbidities like cardiac valvular lesions, pericarditis, and pulmonary fibrosis should be ruled out in rheumatoid arthritis.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

Q50. Answer: B — Rheumatoid arthritis and Scleroderma

Pulmonary fibrosis is mentioned as a comorbidity of both rheumatoid arthritis (preoperative evaluation) and scleroderma (multiorgan involvement).

Topic: RA and Scleroderma | Ref: Ch 25, Page 214